Assignment 1

Department: Computer Engineering & Applications	**Set:** I
Programme: B.Tech	**Branch: CSE**
Subject Name: Computer Network	**Subject Code:** BCSC0008
Year: II	**Semester:** IV

Q1. In a token ring network, the transmission speed is 20 bps, and the propagation speed is 180 m/ms. The 1-bit delay in this network is equivalent to:

Q2. Determine the maximum cable length (in km) for transmitting data at a rate of 400 Mbps in an Ethernet LAN with frames of size 8000 bits. Assume the signal speed in the cable to be 250000 km/s and the link utilization is 50%.

Q3. Consider a token ring LAN of length 12 km and having 40 stations, signal propagation speed is 8 ns/m and data rate is 100 Mbps. An average frame contains 220 bytes. Delay occurred at each station is equivalent to 10-bit delay. What is the utilization of token ring approximately?

Q4. Assume six devices are arranged in a mesh topology. How many cables are needed? How many ports are needed for each device?

Q5. For each of the following four networks, discuss the consequences if a connection fails.
a. Five devices arranged in a mesh topology
b. Five devices arranged in a star topology (not counting the hub)
c. Five devices arranged in a bus topology
d. Five devices arranged in a ring topology

Q6. Performance is inversely related to delay. When you use the Internet, which of the following applications are more sensitive to delay?
a. Sending an e-mail
b. Copying a file
c. Surfing the Internet

Q7. When a party makes a local telephone call to another party, is this a point-to-point or multipoint connection? Explain your answer.

Q8. Match the following to one or more layers of the OSI model:
a. Route determination
b. Flow control
c. Interface to transmission media
d. Provides access for the end user

Q9. Match the following to one or more layers of the OSI model:
a. Format and code conversion services
b. Establishes, manages, and terminates sessions
c. Ensures reliable transmission of data
d. Log-in and log-out procedures
e. Provides independence from differences in data representation

Q10. Suppose a computer sends a frame to another computer on a bus topology LAN. The physical destination address of the frame is corrupted during the transmission. What happens to the frame? How can the sender be informed about the situation?

Q11. Suppose a computer sends a packet at the network layer to another computer somewhere in the Internet. The logical destination address of the packet is corrupted. What happens to the packet? How can the source computer be informed of the situation?

Q12. Suppose a computer sends a packet at the transport layer to another computer somewhere in the Internet. There is no process with the destination port address running at the destination computer. What will happen?

Q13.We have a channel with a 1 MHz bandwidth. The SNR for this channel is 63. What is the appropriate bit rate and signal level?

Q14. Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this calculate the channel capacity.

Q15. Consider a 3 Mbps token ring LAN and frame size is of 200 byte. If the ring latency is 150ms, then what will be the effective data rate of the LAN?